



(12) **United States Patent**
Mendes et al.

(10) **Patent No.:** **US 12,404,976 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **METHOD AND SYSTEM FOR PRESSURE RELIEF IN A MULTI CHAMBER VESSEL**

2201/0109; F17C 2201/0166; F17C 2205/0332; F17C 2209/22; F17C 2203/069; B65D 83/70; B65D 51/1638; Y10T 137/1744; Y10T 137/1714; Y10T 137/1752

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See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/584,360**

(22) Filed: **Feb. 22, 2024**

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(65) **Prior Publication Data**

US 2024/0191844 A1 Jun. 13, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/727,251, filed on Apr. 22, 2022, now Pat. No. 11,953,154.

(51) **Int. Cl.**
F17C 13/12 (2006.01)
F17C 1/00 (2006.01)
F17C 13/04 (2006.01)

(52) **U.S. Cl.**
CPC **F17C 13/12** (2013.01); **F17C 1/00**
(2013.01); **F17C 13/04** (2013.01);
(Continued)

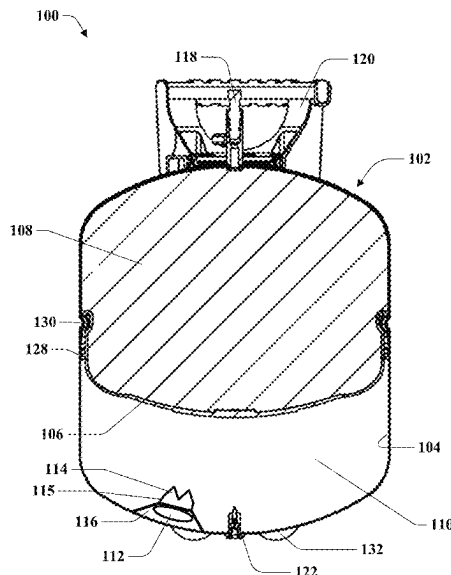
(58) **Field of Classification Search**

CPC .. F17C 13/12; F17C 1/00; F17C 13/04; F17C

(57) **ABSTRACT**

A vessel includes a body that defines an interior space. The body includes a first metal surface, a second metal surface, and a non-metallic material crimped between the first metal surface and the second metal surface at a joint. The vessel further includes a pressure relief device coupled to the body. The pressure relief device is configured to vent contents of the vessel to an exterior of the body. Upon exposure of the vessel to a temperature for a period of time, the second metal surface is configured to expand or bend to create a pressure relief route from the interior space to the pressure relief device, between the first metal surface and the second metal surface, and/or the non-metallic material is configured to melt to create the pressure relief route.

20 Claims, 8 Drawing Sheets



(52) U.S. Cl.

CPC *F17C 2201/0109* (2013.01); *F17C 2201/0166* (2013.01); *F17C 2205/0332* (2013.01); *F17C 2209/22* (2013.01)

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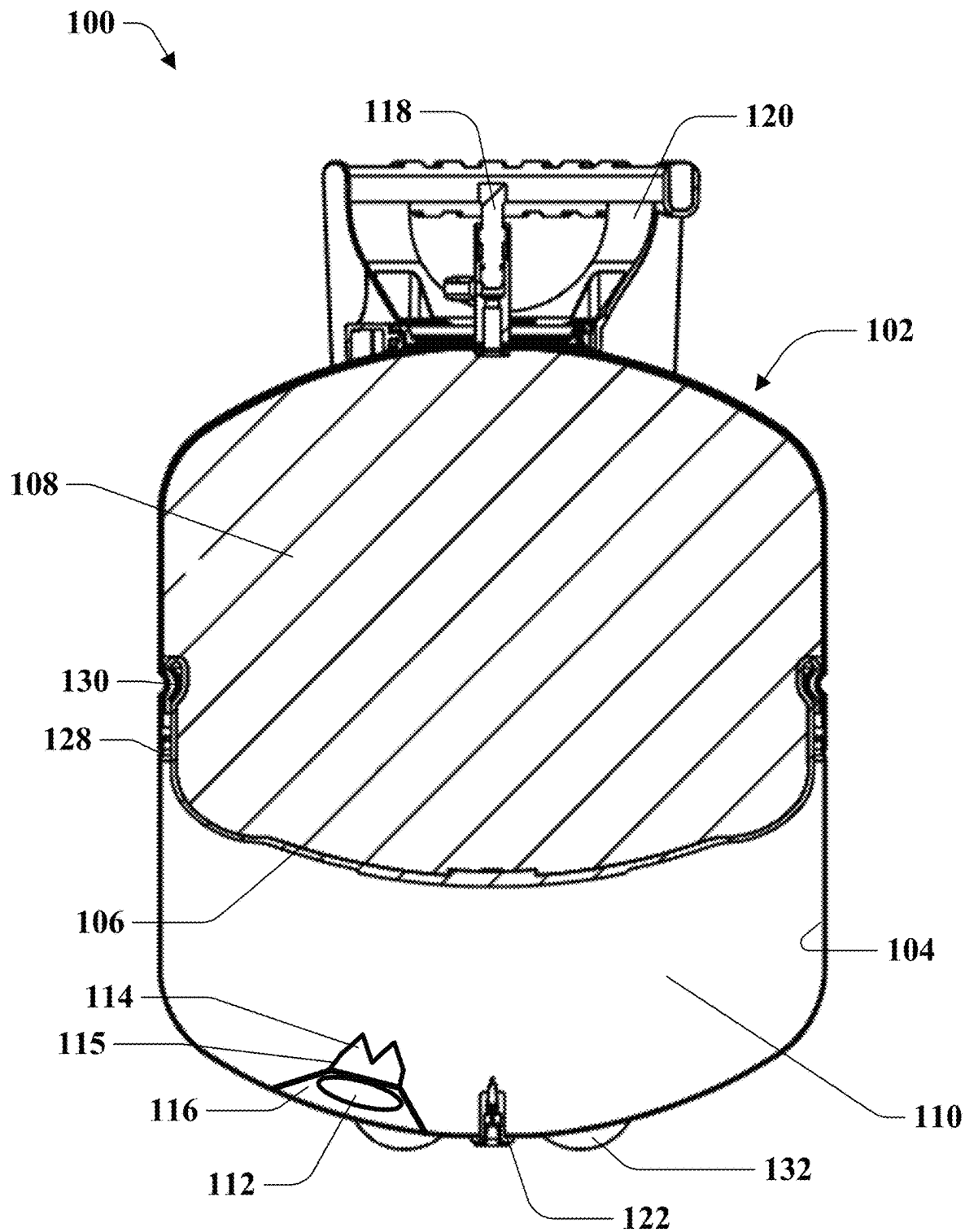


FIG. 1

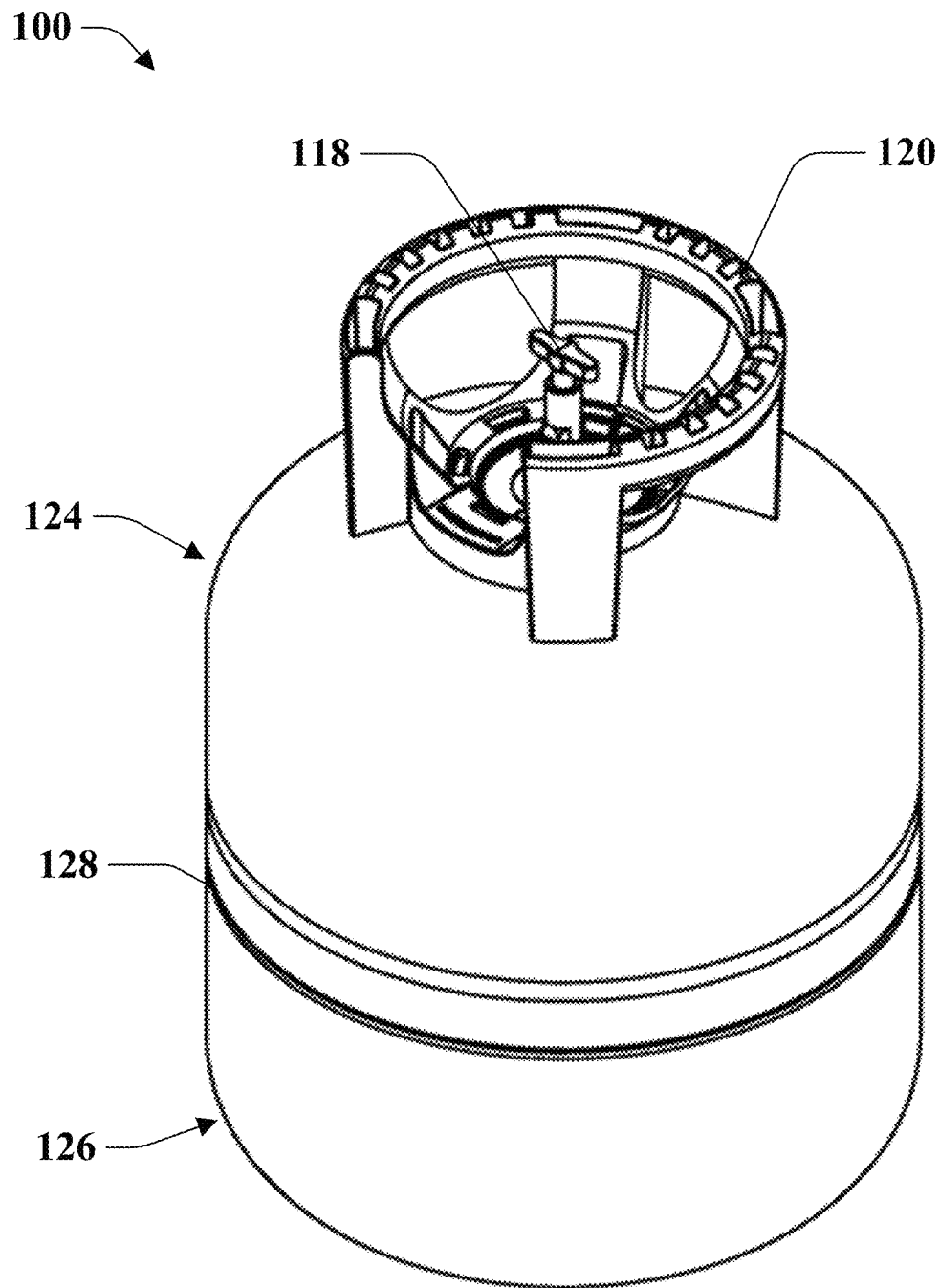
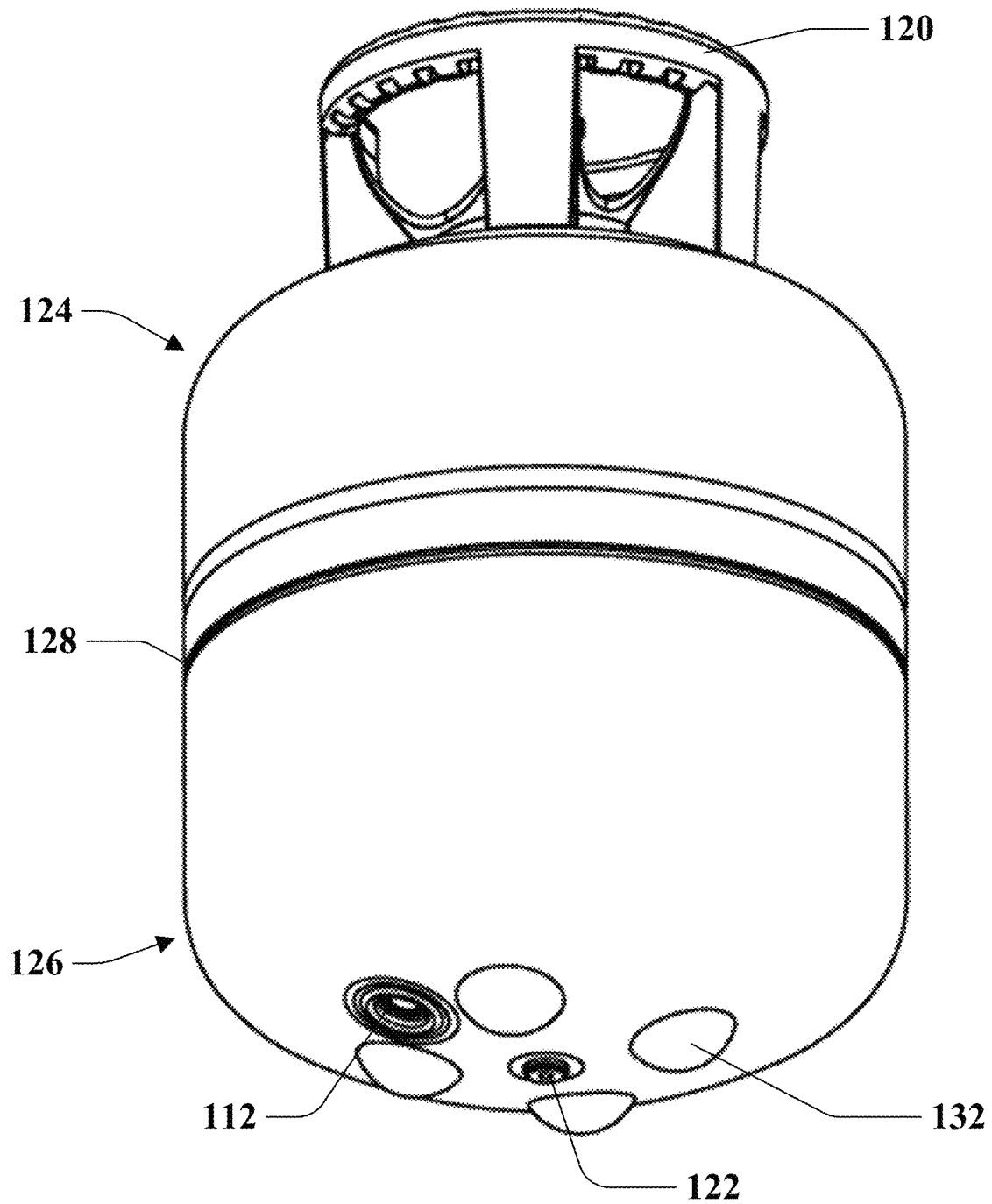


FIG. 2

**FIG. 3**

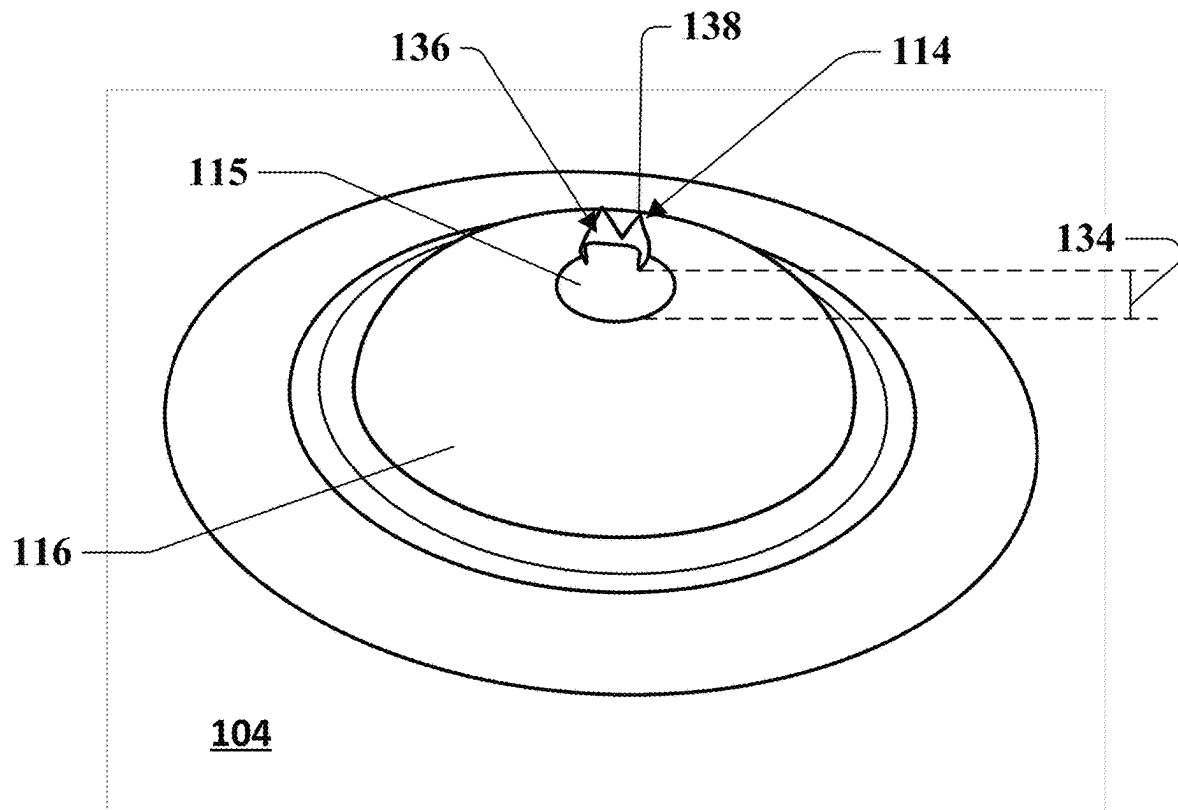


FIG. 4A

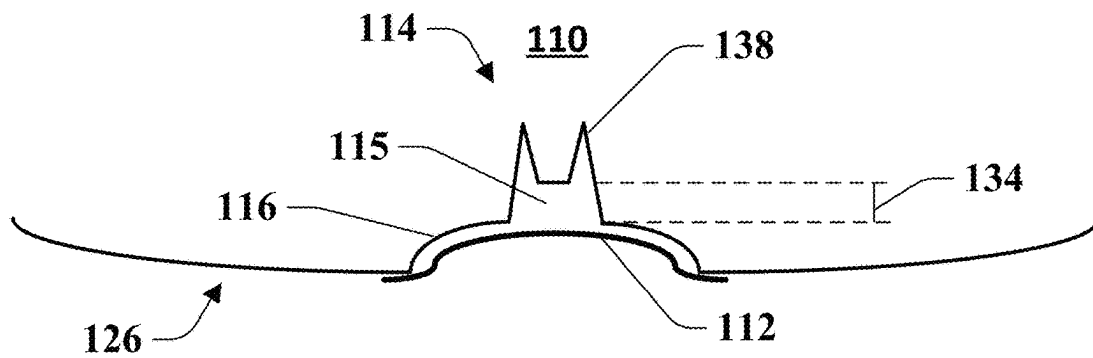


FIG. 4B

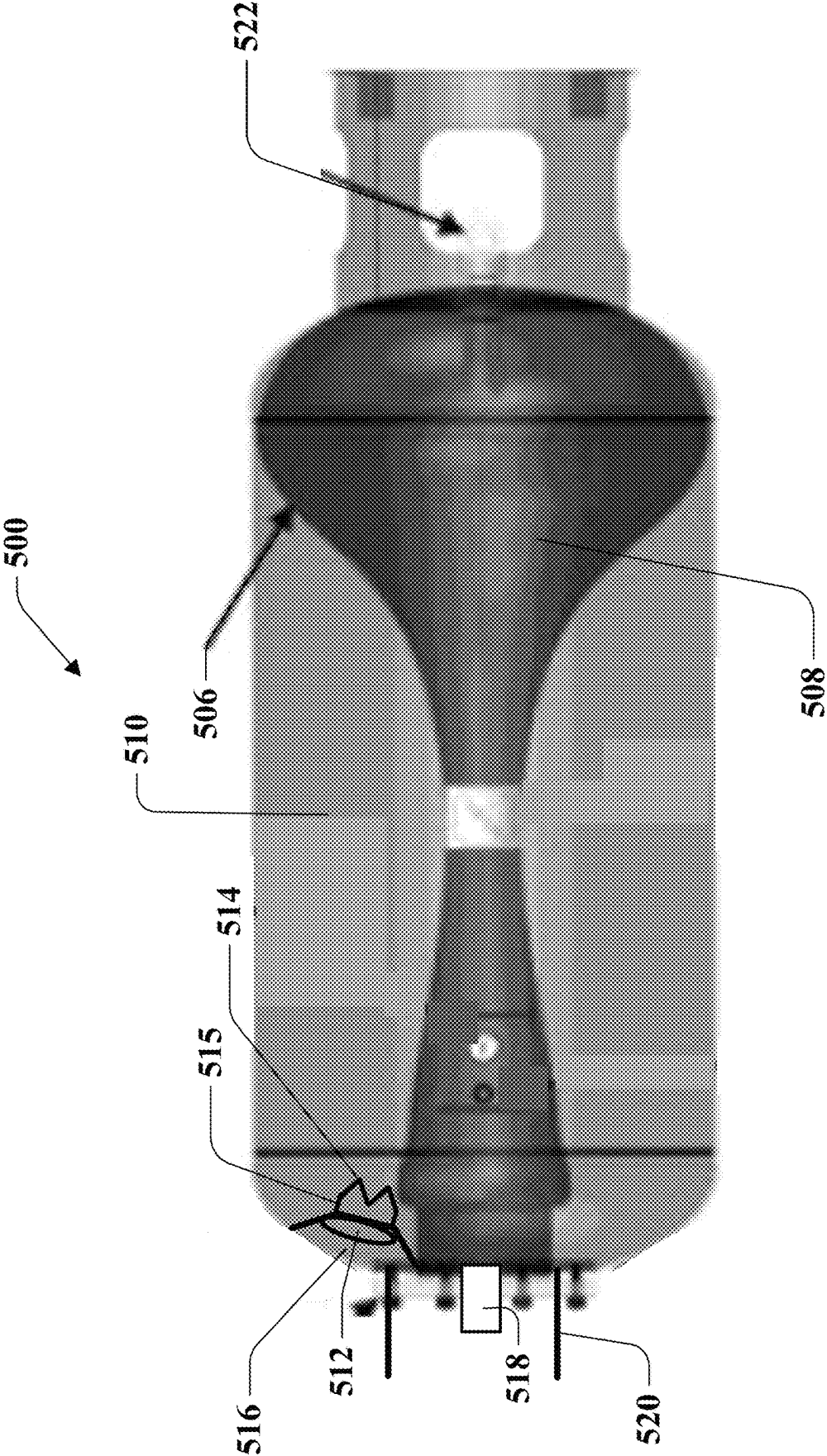


FIG. 5A

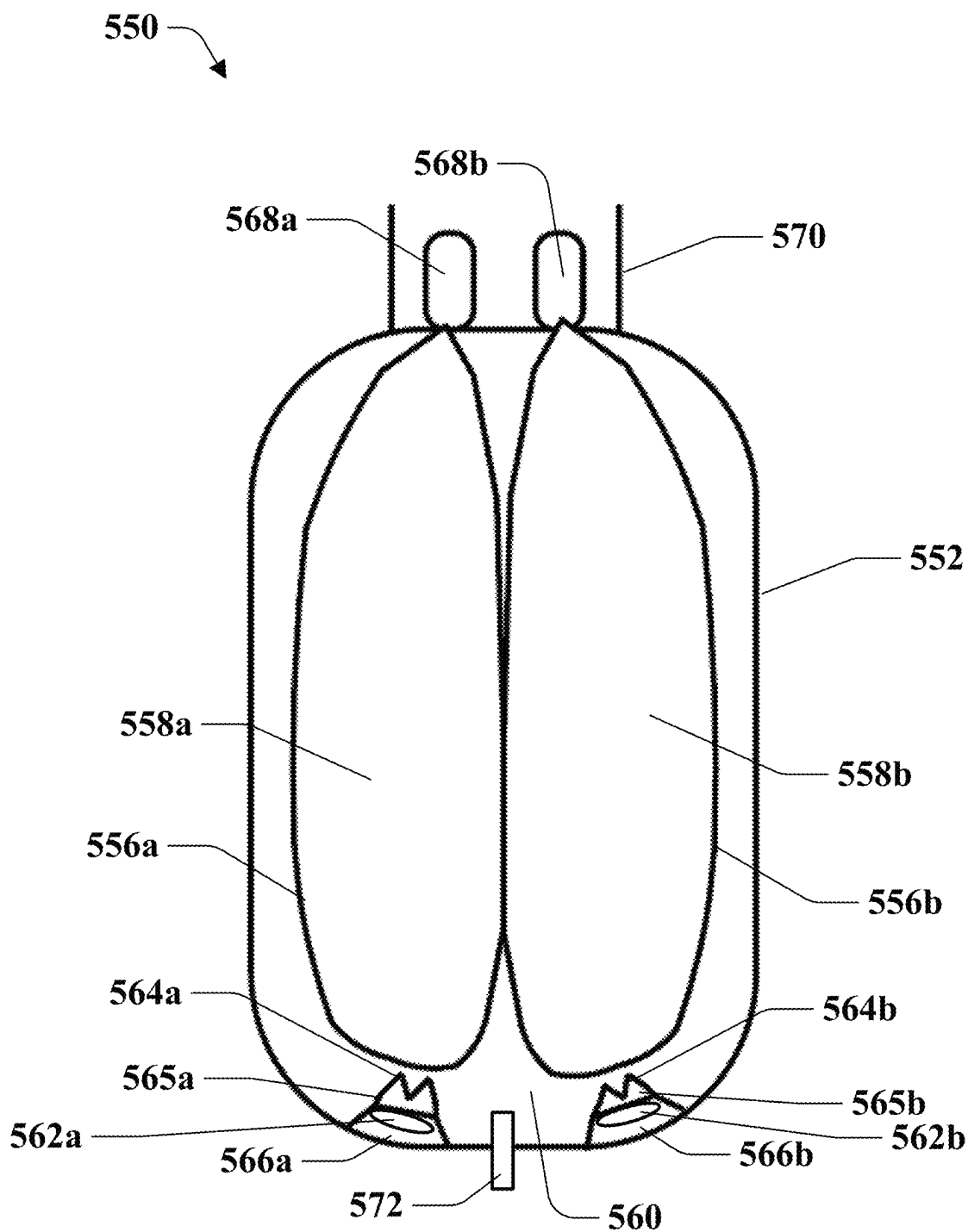
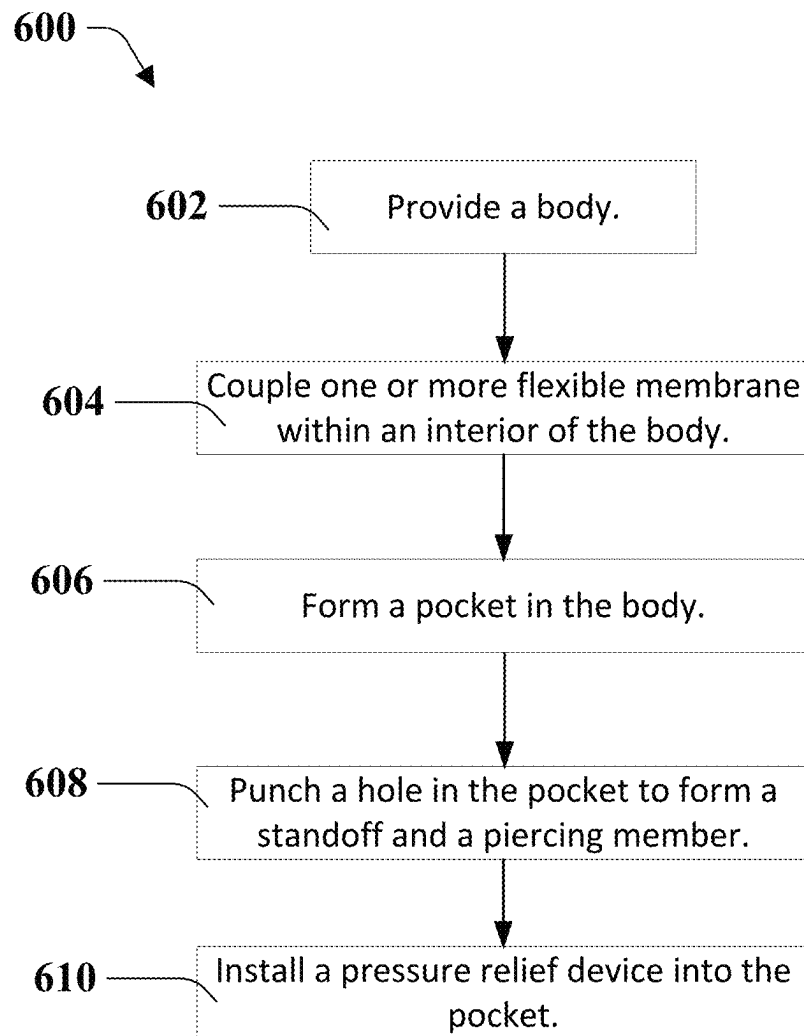
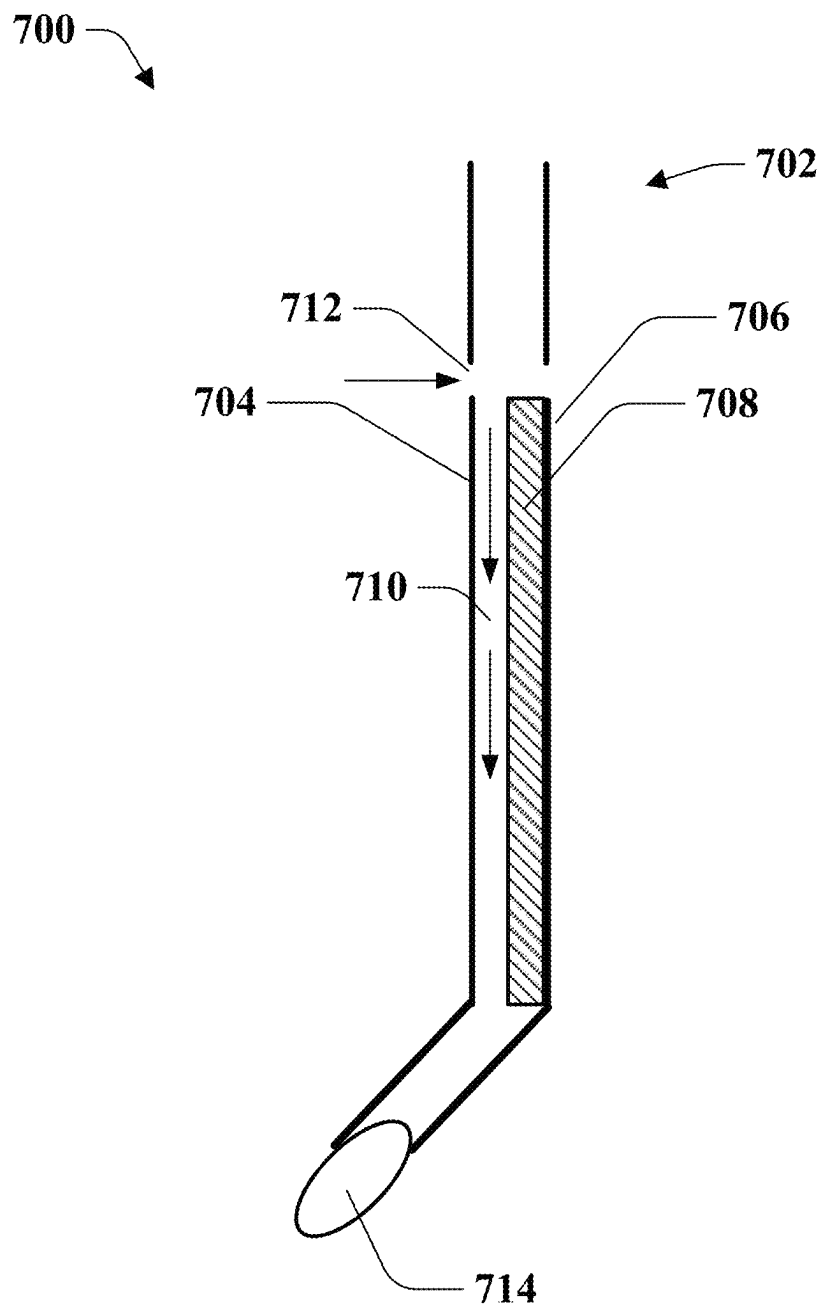


FIG. 5B

**FIG. 6**

**FIG. 7**

1

METHOD AND SYSTEM FOR PRESSURE RELIEF IN A MULTI CHAMBER VESSEL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of and claims the benefit of U.S. Non-provisional application Ser. No. 17/727,251 filed Apr. 22, 2022 and entitled “METHOD AND SYSTEM FOR PRESSURE RELIEF IN A MULTI CHAMBER VESSEL”. The entirety of the aforementioned application is incorporated herein by reference.

BACKGROUND

Vessels can be used for storage and/or dispensing of gasses, liquefied compressed gasses or liquids. Certain vessels can include multiple internal chambers separated by a flexible membrane, for example, a diaphragm, a bladder, or a bag. In vessels where one of the chambers does not have a means for relieving excess pressure, dangerous over-pressure situations can occur. For example, high pressure situations may occur due to a fire, excess heat, or overfilling. Such high-pressure situations may cause dangerous consequences such as explosions and catastrophic ruptures creating projectiles, which may injure people and damage equipment.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key factors or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

One or more techniques and systems described herein can be utilized to provide a vessel that includes a body having an interior surface that defines an interior space. A flexible membrane such as a diaphragm, a bag, or a bladder can be located within the interior space of the vessel. The interior space can include a first chamber at least partially defined by the flexible membrane, and a second chamber at least partially defined by the flexible membrane and a portion of the interior surface of the body. A pressure relief device can also be coupled to the body. The pressure relief device is configured to vent contents of the second chamber to an exterior of the body when the second chamber reaches a first predefined pressure. The flexible membrane is configured to tear or puncture when the first chamber reaches a second predefined pressure that is less than or equal to the first predefined pressure. In this manner, dangerously high pressures in the first chamber can be prevented.

In one embodiment, the vessel includes a piercing member extending into the second chamber. The piercing member is designed and positioned to puncture the flexible membrane when the first chamber reaches the second predefined pressure.

In another embodiment, the flexible membrane is constructed and sized such that the flexible membrane rips or tears upon the first chamber reaching the second predefined pressure.

In another aspect, the vessel can be constructed with walls that include one or more non-metallic materials sandwiched between two metallic layers. In a situation where excess heat or flame is applied to the vessel, thermal expansion of the metallic layers and/or contraction or softening of the one or

2

more non-metallic layers create a pressure relief channel. The pressure relief channel allows for contents from the interior of the vessel to vent from the interior through the pressure relief channel, and through the pressure relief device to the exterior of the vessel.

To the accomplishment of the foregoing and related ends, the following description and drawings set forth certain illustrative aspects and implementations. These are indicative of but a few of the various ways in which one or more aspects may be employed. Other aspects, advantages and novel features of the disclosure will become apparent from the following detailed description when considered in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a component diagram illustrating an example embodiment of a vessel, where one or more techniques and/or one or more systems described herein may be implemented;

FIG. 2 is a top perspective view of an example embodiment of a vessel;

FIG. 3 is a bottom perspective view of an example embodiment of a vessel;

FIG. 4A is an enlarged view of an exemplary piercing member;

FIG. 4B is a cross sectional view of an exemplary pocket;

FIG. 5A is a component diagram illustrating another example embodiment of a vessel;

FIG. 5B is a component diagram illustrating another example embodiment of a vessel;

FIG. 6 is a flow chart depicting an exemplary method of constructing a vessel; and

FIG. 7 is a cross-sectional view of an example embodiment of a vessel wall.

DETAILED DESCRIPTION

The claimed subject matter is now described with reference to the drawings, wherein like reference numerals are generally used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the claimed subject matter. It may be evident, however, that the claimed subject matter may be practiced without these specific details. In other instances, structures and devices are shown in block diagram form in order to facilitate describing the claimed subject matter.

A vessel, as used herein, is defined as any container capable of storing and/or dispensing contents contained within. By way of example and not limitation, a vessel, as used herein, may be a pressure vessel.

A vessel can include two or more chambers separated by a flexible membrane such as a diaphragm, a bag, or a bladder. The vessel can include a pressure relief device located on the body of the vessel. This pressure relief device can function to relieve/vent pressure from one chamber that is in direct fluid communication with the pressure relief device. However, the pressure relief device is unable to relieve/vent pressure from the other one or more chambers while the flexible membrane is intact. Rather, the vessel can include one or more means to tear or puncture the flexible membrane so that the pressure of the one or more other chambers may also be relieved and vented to the exterior of the vessel through the pressure relief device.

Turning now to FIG. 1, a vessel **100** is shown. The vessel **100** can be, for example, a storage tank or an expansion tank such as a diaphragm tank or a bladder tank, among others. Vessel **100** includes a body **102** that can be constructed of any material chosen using sound engineering judgment. By way of example, and not limitation, the body **102** can be constructed using one or more of metal, such as steel or aluminum, carbon fiber, glass fiber, or a polymer such as high density polyethylene. The body **102** includes an interior surface **104** that defines an interior space of the vessel **100**.

The vessel **100** can further include a flexible membrane **106** located within the interior space of the vessel **100**. The flexible membrane **106** can form distinct chambers within the interior space of the vessel **100**. In one embodiment, a first chamber **108** can be at least partially defined by the flexible membrane **106**. A second chamber **110** can be at least partially defined by the flexible membrane **106** and a portion of the interior surface **104** of the body **102**. It should be appreciated that certain embodiments of the vessel **100** can include additional flexible membranes and/or additional chambers. During normal operation of the vessel **100**, the flexible membrane **106** causes the first chamber **108** and the second chamber **110** to be fluidly isolated from each other such that there is no fluid communication between the first chamber **108** and the second chamber **110** within the interior space of the vessel **100**. By way of example, and not limitation, the flexible membrane **106** can be constructed from an elastomer (e.g. ethylene propylene diene monomer (EPDM), butyl, nitrile, neoprene or silicone rubber), a film such as a polyester film (e.g. biaxially-oriented polyethylene terephthalate (BoPET)), which may be a single-ply film or a multi-ply film, or a foil.

The vessel **100** can further include a pressure relief device **112** coupled to the body **102**. The pressure relief device **112** can be, for example, a pressure relief valve, a burst disc, or any other pressure-activated relief mechanism. The pressure relief device **112** can be located on the vessel **100** body **102** on the perimeter of the second chamber **110**, and can be configured to vent the contents of the second chamber **110** to the exterior of the body **102** when the second chamber **110** reaches a first predefined pressure. The first predefined pressure can be selected to prevent excess pressure from building within the vessel **100** and damaging the vessel **100** or causing a dangerous explosion or projectile. In one embodiment, the first predefined pressure can be selected to be greater than or equal to 325 pounds per square inch (psi). In another embodiment, the first predefined pressure can be selected to be greater than or equal to 112 psi. In one embodiment, when the pressure within the second chamber **110** reaches the first predefined pressure, the pressure relief device **112** opens to vent the contents of the second chamber **110** to the exterior of the vessel **100** and cause the second chamber **110** to have ambient pressure. It should be appreciated that the first predefined pressure can be a particular pressure, a particular pressure with a tolerance, or a range of pressures.

In certain embodiments, the vessel can include a piercing member **114**. The piercing member **114** extends into the second chamber **110** towards the flexible membrane **106**. In different embodiments, the piercing member **114** can extend directly from an interior surface **104** of the body **102** into the second chamber **110**, extend directly from the pressure relief device **112**, extend from a standoff **115** that separates the piercing member **114** from the pressure relief device **112** by a distance, or extend from another component of the vessel

100. The piercing member **114** can be, for example, one or more razors, burrs, serrated surfaces, needles, or other sharp or rough surface.

The piercing member **114** is configured to puncture the flexible membrane **106** when the first chamber **108** reaches a second predefined pressure. In certain embodiments, the second predefined pressure is less than or equal to the first predefined pressure at which the pressure relief device **112** operates to vent the second chamber **110**. During normal operation when the pressure relief device **112** is not activated, the first chamber **108** and the second chamber **110** share a common pressure. If the flexible membrane **106** remains intact long after the pressure relief device **112** opens and begins venting the second chamber **110**, there is a risk that the flexible membrane **106** can continue to expand and seal or significantly restrict the pressure relief device **112**, thereby preventing the vessel **100** from being properly vented. In this situation, the dangerous high pressure can build again within the vessel **100**. Accordingly, the flexible membrane **106** and/or piercing member **114** can be designed to cause the flexible membrane **106** to tear or puncture prior to making contact with the pressure relief device **112**. To achieve this end, the second predefined pressure can be chosen to be less than or equal to the first predefined pressure. It should be appreciated that the second predefined pressure can be a particular pressure, a particular pressure with a tolerance, or a range of pressures.

Other features may be included within the vessel **100** to prevent the flexible membrane **106** from sealing the pressure relief device **112**. For example, a portion of the interior surface **104** within the second chamber **110**, a surface of the flexible membrane **106**, or both can include one or more projections. These projections can include one or more of channels, bumps, ridges, or knurls, among others. The projections can prevent a complete seal from being formed in the event that the flexible membrane **106**, or a portion thereof, happens to make contact with the pressure relief device **112** or any aperture that provides access to the pressure relief device **112**.

In certain embodiments, the vessel **100** can include a pocket **116** formed in the body **102** of the vessel **100**. The pocket **116** can be formed in the body **102** such that the pocket extends inwards into the second chamber **110**. For example, the pocket **116** can include a convex shape in the body **102**, indented within the second chamber **110**. The standoff **115** can extend from the pocket **116** into the second chamber **110**, and the piercing member **114** can extend from the standoff **115** into the second chamber **110**, as shown in FIG. 1. The pressure relief device **112** can be installed into the pocket **116**. For example, the pressure relief device **112** can fit into and be mounted on an exterior of the indent formed by the pocket **116**.

As an example of operation of an embodiment of the vessel **100**, the pressure increases within the vessel **100**. The vessel's **100** pressure increase causes the pressure to increase in both the first chamber **108** and the second chamber **110**. As the pressure increases in the first chamber **108**, the flexible membrane **106** expands into space previously occupied by the second chamber **110**. It should be appreciated that the body **102** maintains its shape as the flexible membrane **106** expands. The flexible membrane **106** continues to expand as the pressure of the first chamber **108** increases towards the second predefined pressure, and the flexible membrane **106** approaches the piercing member **114**. As the pressure of the first chamber **108** reaches the second predefined pressure, the flexible membrane **106** contacts the piercing member **114** with enough force to be

5

punctured by the piercing member 114. Simultaneously, or soon after the flexible member 106 is punctured by the piercing member 114, the second chamber 110 reaches the first predefined pressure, which causes the pressure relief device 112 to open.

In one embodiment, the flexible membrane 106 is configured to be punctured and/or tear at a second predefined pressure, but only after the pressure relief device 112 opens at the first predefined pressure. In this situation, both the first chamber 108 and the second chamber 110 can reach the first predefined pressure. The pressure relief device 112 is configured to open at the first predefined pressure, which causes the pressure of the second chamber 110 to drop to 0 psi. Simultaneously with the decrease in pressure of the second chamber 110, the flexible membrane 106 expands, causing the pressure of the first chamber 108 to also decrease to less than the first predefined pressure at which the pressure relief device 112 opened. Over time, as pressure continues to build in the first chamber 108, the flexible membrane 106 continues to expand and contacts the piercing member 114 with enough force to puncture the flexible membrane 106.

In one embodiment, the flexible membrane 106 is configured to tear when the first chamber reaches the second predefined pressure. The strength, thickness, and type of material can be selected to ensure that the flexible membrane 106 fails when the first chamber 108 reaches the second predefined pressure. It should be appreciated that while this embodiment may not require a piercing member 114 to puncture the flexible membrane 106, a piercing member 114 may still be included as a backup to ensure that the flexible membrane 106 is punctured in the event that the flexible membrane 106 does not tear as expected.

The vessel 100 can further include components such as a valve 118 coupled to the vessel 100, and in fluid communication with the first chamber 108. A shroud 120 can also encircle the valve 118 to provide protection to the valve 118 and also provide handles for grasping and lifting the vessel 100. A propellant valve 122 can also be coupled to the body 102 to provide a selective fluid communication between the interior space of the vessel 100 and a source of a propellant used to create a pressure differential between the second chamber 110 and the exterior of the vessel 100 so that the contents of the first chamber 108 can be expelled out of the valve 118. In one embodiment, the propellant valve 122 provides selective fluid communication between the second chamber 110 and a source of a propellant. In this embodiment, the propellant valve 122 can be utilized by a user for filling the second chamber 110 with a pressurized gas such as pressurized air, nitrogen, carbon dioxide, or compressed liquified gas such as propane, butane or refrigerant, among others. The propellant valve 122 can extend into the second chamber 110. In certain embodiments, the propellant valve 122 can include a piercing member 114 extending therefrom into the second chamber 110 towards the flexible membrane 106. In these embodiments, the propellant valve 122 can include the sole piercing member 114, or it could also include an auxiliary piercing member 114 in addition to a primary piercing member 114 located elsewhere on the vessel 100 (e.g. on the pressure relief device 112, pocket 116, or standoff 115). In another embodiment, a portion of the propellant valve 122 that extends into the second chamber 110 is a piercing member 114.

With further reference to FIG. 2 and FIG. 3, the body 102 can be constructed from a first portion 124 and a second portion 126 that are connected by, for example, welding, brazing, crimping, or flange. The connection of the first portion 124 and the second portion 126 can create a seam

6

128. In certain embodiments, the flexible membrane 106 can be crimped to at least one of the first portion 124 or the second portion 126 of the vessel 100 body 102. In certain embodiments, the flexible membrane 106 is crimped between the first portion 124 or the second portion 126, and a crimping ring 130, which extends around the inner circumference of the vessel 100 body 102. In other embodiments, the body 102 of vessel 100 can be seamless.

In certain embodiments, both of the first portion 124 and the second portion 126 include a convex shell and a cylindrical sidewall portion. The first portion 124 can be configured to receive the valve 118, which provides selective fluid communication between the first chamber 108 and the exterior of the body 102. The valve 118 can be used either for filling the vessel 100, dispensing contents of the vessel 100, or both. A shroud 120 can also be mounted to the first portion 124. The second portion 126 can include one or more of the pressure relief device 112, the piercing member 114, and the propellant valve 122. The second portion 126 can further include a stand 132 such as a footing or plurality of dimples for feet on which the vessel 100 can stand. The stand 132 provides balance and support for the vessel 100. In one embodiment, the pressure relief device 112 and the piercing member 114 are both located on the convex shell of the second portion 126. Specifically, any weldments and openings for the pressure relief device 112 and/or the piercing member 114 are both located within the innermost 80% portion of the radius of the convex shell, if located on the convex top portion of the second portion 126. There are no location limitations, however, if pressure relief device 112 and/or piercing member 114 are on the side walls of the second portion 126. It should be appreciated that the vessel 100 can be operable while positioned in any orientation, including with the first portion 124 oriented upwards and the second portion 126 oriented downwards, the first portion 124 oriented downwards and the second portion 126 oriented upwards, or the vessel 100 oriented on its side such that the first portion 124 and the second portion 126 extend along a common horizontal plane.

Turning now to FIGS. 4A and 4B, an exemplary piercing member 114 is shown. The piercing member 114 can extend from the standoff 115, which separates the piercing member 114 from the pressure relief device 112 by a vertical distance 134 perpendicular to the interior surface 104 of the vessel 100. By separating the piercing member 114 from the pressure relief device 112 by the distance 134, the likelihood of the flexible membrane 106 contacting or sealing the pressure relief device 112 while the pressure relief device 112 is venting the second chamber 110 to the exterior of the vessel 100 is reduced. The standoff 115 can extend from the pocket 116, which at least partially surrounds the pressure relief device 112. In one embodiment, the pocket 116 includes a convex surface extending into the second chamber 110, which also reduces the likelihood of the flexible membrane 106 contacting or sealing the pressure relief device 112 while the pressure relief device 112 is venting the second chamber 110 to the exterior of the vessel 100. As shown in the cross-sectional view of FIG. 4B, the pressure relief device 112 can be mounted at least partially within the indent formed by the pocket 116, on an exterior of the body 102.

The piercing member 114 and the standoff 115 can form an aperture 136 that defines a passageway that provides fluid communication between the pressure relief device 112 and the second chamber 110. In one embodiment, the aperture 136 is at least 10 millimeters in diameter to reduce the likelihood of the flexible membrane 106 from restricting or

7

sealing the pressure relief device 112 when venting the first chamber 108 and/or second chamber 110 to the exterior of the vessel 100. The piercing member 114 can include a plurality of piercing points 138 surrounding the aperture 136. Each of the plurality of piercing points 138 can be separated from each other by a space. In certain embodiments, the piercing member 114 can be formed from the standoff 115. In these embodiments, the piercing member 114, including any piercing points 138, project directly from the standoff 115.

FIG. 5A depicts another exemplary embodiment of a vessel 500. It should be appreciated that the construction of vessel 500 can be similar to that of vessel 100 and include two portions connected together at a seam, or alternatively be seamless. The vessel 500 includes a flexible membrane 506. The flexible membrane 506 is in the form of a bladder or bag that completely defines the first chamber 508. The flexible membrane 506 can be attached to the valve 518 within the vessel 100 or at the junction of the valve 518 and the body 502. The second chamber 510 exists as the area outside of the flexible membrane 506, defined by the inner surface 504 of the vessel 500 and the outside of the flexible membrane 506. Similarly as the embodiments described above with respect to vessel 100, vessel 500 can include a pressure relief device 512 in the body 502 of the vessel 500, and a piercing member 514 that can extend from a standoff 515 that spaces the piercing member 514 from the pressure relief device 512.

Similar to vessel 100, vessel 500 can include a pocket 516 formed in the body 502. The standoff 515 can extend from the pocket 516 into the second chamber 510, and the piercing member 514 can extend from the standoff 515. Vessel 500 can further include a valve 518 and a propellant valve 522 that serve the same functions as in vessel 100 as described with respect to FIG. 1. The valve 518 can be at least partially surrounded by a shroud 520. It should be appreciated that the propellant valve 522 may also be at least partially surrounded by a second shroud. Propellant valve 522 can be located on an end of the vessel 500 (as shown) or it can alternatively be located on a side wall of the vessel 500. If the propellant valve 522 is mounted to either end of the vessel 500, it may be mounted within the innermost 80% portion of the radius of the convex shell that makes up an end of the vessel 500. Vessel 500 can further include a stand (not shown in FIG. 5A) such as a footing or a plurality of dimples for feet on which the vessel 500 can stand. The stand can be included on either end of the vessel 500, and any valve or shroud (e.g. valve 518, shroud 520, propellant valve 522) may be sized, positioned, or omitted accordingly to allow the stand to function without obstruction. For example, a stand can be included near the propellant valve 522. In this case, the propellant valve 522 can be sized to not extend past the stand, and a shroud around the propellant valve 522 can be omitted. In another example, a shroud (e.g. shroud 520, or a shroud around the propellant valve) can also function as a stand.

FIG. 5B depicts another exemplary embodiment of a vessel 550. Vessel 550 includes a first flexible membrane 556a that at least partially defines a first chamber 558a, and a second flexible membrane 556b that at least partially defines a second chamber 558b. The vessel 550 also includes a third chamber 560 at least partially defined by the body 552 of the vessel 550. Vessel 550 can include a first valve 568a that provides selective fluid communication into and out of the first chamber 558a and a second valve 568b that provides selective fluid communication into and out of the second chamber 558b. In certain embodiments, the first

8

valve 568a and the second valve 568b are integrated into a single valve body. A shroud 570 can surround both the first valve 568a and the second valve 568b, or the vessel 550 can include a shroud 570 for each of the first and second valves 568a, 568b. Vessel 550 can include at least one of a first piercing member 564a, a first standoff 565a, and/or a first pocket 566a, which correspond to the first flexible membrane 556a, and function in the same manner as described in the other embodiments above. For example, the first piercing member 564a can be arranged and configured to puncture the first flexible membrane 556a when the first chamber 558a reaches a particular pressure. Further, vessel 550 can include at least one of a second piercing member 564b, a second standoff 565b, and/or a second pocket 566b, which correspond to the second flexible membrane 556b, and function in the same manner as described in the other embodiments above. For example, the second piercing member 564b can be arranged and configured to puncture the second flexible membrane 556b at a particular pressure of the second chamber 558b. The vessel 550 can include at least one pressure relief device 562a, 562b to selectively vent pressure from the third chamber 560 to the exterior of the vessel 550. For example, a first pressure relief device 562a can be mounted within the first pocket 566a and/or the second pressure relief device 562b can be mounted within the second pocket 566b.

Turning now to FIG. 6, a method for constructing a vessel 100 is depicted. At reference numeral 602, a body 102 is provided. At reference numeral 604, one or more flexible membranes 114 are coupled within an interior of the body 102 to create at least a first chamber 108 and a second chamber 110 within the body 102. For example, the flexible membrane 106 can be crimped to at least one of the first portion 124 or the second portion 126 with a crimping ring 130, or the flexible membrane 106 can attach to a valve 118 coupled with the body 102. At reference numeral 606, a pocket 116 is formed in the wall of the body 102. In one embodiment, the pocket 116 can be formed to include a convex shape indented within the second chamber 110. In certain embodiments, the pocket 116 can be formed using a die. At reference numeral 608, a hole is punched in the pocket 116 to form a standoff 115 and a piercing member 114 that includes an aperture 136 and piercing points 138 surrounding the aperture 136 and projecting from the standoff 115. The aperture 136 can act as an entrance to a passageway that provides fluid communication between a pressure relief device 112 and the second chamber 110. For example, in an embodiment where the pocket 116 has a convex shape, the hole is punched from the inside of the convex shape (corresponding to the exterior of the vessel 100) outwards so that the piercing points 138 extend outwards from the convex surface of the pocket 116. In other words, the punching force is directed inwards towards the interior of the body 102. In one embodiment, the aperture 136 is created to be at least 10 millimeters in diameter. At reference numeral 610, a pressure relief device 112 is installed into the pocket 116 such that the pressure relief device 112 provides selective fluid communications between the second chamber 110 and an exterior of the body 102 when the pressure relief device 112 is activated at the first predefined pressure.

Turning now to FIG. 7, an exemplary embodiment of a vessel body 702 is shown. Vessel body 702 can include a first metal surface 704 and a second metal surface 706 positioned substantially parallel to each other. In certain embodiments, the first metal surface 704 and the second metal surface 706 can be made of steel. The body 702 can

further include a layer of non-metallic material **708** between the first metal surface **704** and the second metal surface **706**. The non-metallic material **708** can include a single layer or multiple layers, and can be constructed of a single material or multiple materials. In an embodiment, the non-metallic material **708** is a portion of a flexible membrane **106** such as a diaphragm, a bag, or a bladder. In certain embodiments, the non-metallic material **708** can melt when the body **702** is exposed to flames or extreme heat. By way of example and not limitation, the non-metallic material **708** can be made of one or more of polypropylene or a rubber, including Butyl, EPDM rubber, or a blend of Butyl and EPDM Rubber. In one embodiment, the non-metallic material **708** includes a first layer made of polypropylene and a second layer made of EPDM rubber. The material of the first metal surface **704**, the second metal surface **706**, and the non-metallic material **708** can be chosen and configured such that when the body **702** is exposed to heat, the first and second metal surfaces **704**, **706** expand due to their coefficient of thermal expansion. The expansion or bending action of the first and second metal surfaces **704**, **706** creates at least one pressure relief route **710** in between at least one of the first metal surface **704** and the non-metallic material **708**, or the second metal surface **706** and the non-metallic material **708**.

The first metal surface **704** can form an interior of a vessel body **702** and the second metal surface **706** can form an exterior of the vessel body **702**. The non-metallic material **708** can be compressed in between the first metal surface **704** and the second metal surface **706**. In certain embodiments, the non-metallic material **708** can be compressed to less than 50% of its uncompressed thickness. In still other embodiments, the non-metallic material **708** can be compressed to less than 20% of its uncompressed thickness. In certain embodiments, the thickness of the non-metallic material can be approximately 0.26 inches when uncompressed, and as thin as 0.05 inches when in a compressed state between the first metal surface **704** and the second metal surface **706**. The first metal surface **704** can be thicker than the second metal surface **706**. During normal operation when there is no fire or excessive heat present, there exists a seal between the first metal surface **704** and the non-metallic material **708** such that the pressure relief route **710** is non-existent.

In certain embodiments, the first surface **704** can have a curved shape. In this embodiment, the non-metallic material **708** can have a corresponding curved shape to create a seal during normal operation. The non-metallic material **708** can also have a curved shape that is a mirrored image of the curved shape of the first surface **704**. The first metal surface **704** can be designed to be more rigid than the second metal surface **706**. This feature can be accomplished either by designing the first metal surface **704** to have a stronger shape (e.g. curved, arched, concave) than the second metal surface **706**, constructing the first metal surface **704** from a stronger material than the second metal surface **706**, or by constructing the first metal surface to have a greater thickness than the second metal surface **706**. In such a configuration, the second metal surface **706** bends outwards when heated by an external heat source, which causes the curvature of the non-metallic material **708** to relax. The relaxation of the curvature of the non-metallic material **708** loosens the seal between the first metal surface **704** and the non-metallic material **708**, and eventually creates a pressure relief route **710** between the first metal surface **704** and the non-metallic material **708**. The pressure relief route **710** can lead directly to an exterior of the vessel **700** or to a pressure relief device **714**, which can then vent the contents of the vessel **700** to

the exterior of the vessel **700**. It should be appreciated that the non-metallic material **708** can be designed to act as a thermal insulator, which further accentuates a thermal expansion difference between first metal surface **704** and second metal surface **706**.

In one embodiment, the non-metallic material **708** (e.g. a flexible membrane, or rigid liner) is crimped between the first metal surface **704** and the second metal surface **706** at a joint **712**. During normal operation of the vessel **700**, the joint **712** creates a seal so that the contents of the vessel **700** cannot escape from the interior of the vessel **700**. When heat is applied to the body **702** that causes the first and/or second metal surface **704**, **706** to expand and/or bend near the joint **712**, the joint **712** can then open to provide an entrance for the contents of the vessel **700** to escape through the at least one pressure relief route **710**. The pressure relief route **710** can lead directly to an exterior of the vessel **700** or to a pressure relief device **714**, which can then vent the contents of the vessel **700** to the exterior of the vessel **700**.

It should be appreciated that the pressure relief route **710** can be created in various ways. For example, if the vessel **700** body **702** is subject to rapid external thermal activity (e.g. 1200 degrees Fahrenheit in 2 minutes or less), the sealing surfaces between the first and second metal surfaces **704**, **706**, and the non-metallic material **708** is relaxed due to differences in thermal expansion between the materials and/or softening of the non-metallic material **708**, resulting in the opening and formation of the pressure relief route **710**. In another example, slower external thermal activity can cause the non-metallic material **708** to melt, and therefore open and form the pressure relief route **710** between the first metal surface **704** and the second metal surface **706**.

In another embodiment, the vessel **700** body **702** can include a melttable component. In one example, the valve coupled with body **702** and utilized for filling and/or dispensing of contents of the vessel **700** can be constructed of a melttable material. In this manner, if the vessel **700** is exposed to excess external heat, the valve or other melttable component will melt and provide a path for the pressurized contents of the vessel **700** to escape.

Moreover, the word “exemplary” is used herein to mean serving as an example, instance or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Further, “at least one of A and B”, or “at least one of A or B” and/or the like generally means A or B or both A and B. In addition, the articles “a” and “an” as used in this application and the appended claims may generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features, ranges, and acts described above are disclosed as example forms of implementing the claims.

Also, although the disclosure has been shown and described with respect to one or more implementations,

11

equivalent alterations and modifications will occur to others skilled in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure includes all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary implementations of the disclosure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application. Furthermore, to the extent that the terms “includes,” “having,” “has,” “with,” or variants thereof are used in either the detailed description or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

The implementations have been described, hereinabove. It will be apparent to those skilled in the art that the above methods and apparatuses may incorporate changes and modifications without departing from the general scope of this invention. It is intended to include all such modifications and alterations in so far as they come within the scope of the appended claims or the equivalents thereof.

What is claimed is:

1. A vessel, comprising:

a body that defines an interior space, wherein the body comprises:

a first metal surface;

a second metal surface; and

a non-metallic material crimped between the first metal surface and the second metal surface at a joint; and a pressure relief device coupled to the body, wherein the pressure relief device is configured to vent contents of the vessel to an exterior of the body;

wherein at least one of:

the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, to create a pressure relief route between the first metal surface and the second metal surface to provide fluid communication between the interior space and the pressure relief device, or the non-metallic material is configured to melt, contract, or soften, upon exposure of the vessel to the temperature for the period of time, to create the pressure relief route.

2. The vessel of claim 1, wherein upon exposure of the vessel to the temperature for the period of time, the second metal surface is configured to expand or bend such that the joint opens to provide an entrance for the contents of the vessel to escape through the pressure relief route.

3. The vessel of claim 1, wherein the second metal surface forms an exterior surface of the body.

4. The vessel of claim 1, wherein the second metal surface is steel.

5. The vessel of claim 1, wherein the first metal surface is thicker than the second metal surface.

6. The vessel of claim 1, wherein the non-metallic material is compressed to less than 50% of its uncompressed thickness.

12

7. The vessel of claim 6, wherein the non-metallic material is compressed to less than 20% of its uncompressed thickness.

8. The vessel of claim 1, wherein the first metal surface is more rigid than the second metal surface.

9. The vessel of claim 1, wherein the non-metallic material includes a flexible membrane that separates the interior space into a first chamber and a second chamber.

10. The vessel of claim 1, wherein the non-metallic material comprises one or more of polypropylene, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, or a blend of butyl and EPDM rubber.

11. The vessel of claim 10, wherein the non-metallic material comprises a first layer made of polypropylene and a second layer made of butyl rubber or EPDM rubber.

12. The vessel of claim 1, wherein the temperature is 1200 degrees Fahrenheit and the period of time is 2 minutes or less.

13. A vessel, comprising:

a body that defines an interior space, wherein the body comprises:

a first metal surface;

a second metal surface; and

a non-metallic material crimped between the first metal surface and the second metal surface at a joint; and wherein the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, such that the joint opens to provide an entrance for a contents of the vessel to flow from the interior space through a pressure relief route between the first metal surface and the second metal surface.

14. The vessel of claim 13, further comprising:

a pressure relief device coupled to the body, wherein the pressure relief route provides fluid communication between the interior space and an exterior of the body via the pressure relief device.

15. The vessel of claim 13, wherein the second metal surface forms an exterior surface of the body.

16. The vessel of claim 13, wherein the first metal surface is thicker than the second metal surface.

17. The vessel of claim 13, wherein the first metal surface is more rigid than the second metal surface.

18. The vessel of claim 13, wherein the non-metallic material comprises one or more of polypropylene, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, or a blend of butyl and EPDM rubber.

19. The vessel of claim 1, wherein the temperature is 1200 degrees Fahrenheit and the period of time is 2 minutes or less.

20. A vessel, comprising:

a body that defines an interior space, wherein the body comprises:

a first metal surface;

a second metal surface exterior to the first metal surface; and

a non-metallic material crimped between the first metal surface and the second metal surface at a joint, wherein the non-metallic material includes:

a flexible membrane that separates the interior space into a first chamber and a second chamber; and a liner;

wherein the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, such that the joint opens to provide an entrance for a contents of the vessel to flow from the

13

interior space through a pressure relief route between
the first metal surface and the second metal surface.

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14